



Genome-wide identification and characterization of Hsp70 gene family in *Nicotiana tabacum*

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Abstract

Heat shock proteins 70 (Hsp70) constitute a highly conserved protein family of cellular chaperones widely distributed in plants, where they play a fundamental role in response to biotic and abiotic stress. Until now, genome-wide analyses of the Hsp70 gene family have been conducted for some species. However, reports about Hsp70 genes in *Nicotiana tabacum* are scarce. In this study, we systematically conducted genome-wide identification and expression analysis of the Hsp70 gene family in tobacco, including gene structure, classification, evolutionary relationships, promoters, and transcript levels in response to abiotic stress treatments. In all, 61 Hsp70 members were identified and classified into six groups that were mapped onto 18 chromosomes, where most were distributed on both ends of the chromosome. The conserved structures and motifs of NtHsp70 proteins in the same subfamily were highly consistent. At least 15 pairs of *NtHsp70* genes underwent gene duplication by segment and tandem duplications. Most NtHsp70 proteins contained N-terminal hexokinase conserved motifs. Phylogenetic analysis showed that most species expanded according to their own species-specific approach during the evolution of Hsp70s. Tissue-specific expression analysis indicated that all *NtHsp70* genes were involved in at least one or more abiotic stress responses, highlighting the wide participation of *NtHsp70* genes in environmental adaptation. This is the first genome-wide analysis of Hsp70 in *N. tabacum*. These results indicate that each *NtHsp70* member fulfilled distinct functions in response to various abiotic stresses.

Keywords *Nicotiana tabacum* · Heat shock protein 70 · Genome-wide analysis · Synteny analysis · Phylogenetic analysis · Expression profile analysis

Abbreviations

Hsp70 Heat shock protein 70
NtHsp70 Gene in *Nicotiana tabacum*
ABA Absciscic acid

PEG Polyethylene glycol
qRT-PCR Quantitative real-time PCR
Chlo Chloroplast
Cyto Cytosol
Endo Endoplasmic reticulum
Mito Mitochondria

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Introduction

Plants are continuously exposed to a variety of adverse conditions in their natural environment, all changing, and interacting at once with plants. According to the specific stimulus, stress can be categorized as biotic or abiotic stress. Among them, abiotic stress conditions (e.g. salinity, drought, cold, or high temperature) are important factors affecting crop growth and development [1, 2].

Under the influence of global warming, high temperature has become a prominent stress factor among abiotic stress conditions [3]. In order to avoid or cope with various stress, plants have evolved a series of regulatory mechanisms during a long evolutionary history [4]. Heat Shock Proteins (Hsps), firstly identified in the salivary gland chromosomes of *Drosophila larvae* [5], is well known to be specifically and significantly induced in plant cells under abiotic stress [6]. Some subsequent studies have confirmed that Hsps occur in all living organisms. Hsps could be classified into five categories based on their approximate molecular weight: Hsp100/ClpB family, Hsp90 family, Hsp70/DnaK family, chaperonin (Hsp60/GroEL) family, and small heat-shock proteins (sHsp) family [7].

Heat shock proteins are produced in response to high temperature or other forms of stress. In all cases, they have all been reported to act in protecting plants from stress [8]. Hsp70s are a highly conserved protein family of cellular chaperones, and widely distributed in prokaryotes and eukaryotes [9]. Under stress conditions, the Hsp70 family members act as molecular chaperones to catalyze folding of the polypeptide chains to form proteins with spatial structures [10]. The Hsp70 gene family mainly contains three highly conserved domains: the C-terminal domain of about 10 kDa that binds to the substrate, the 15 kDa intermediate domain, and the 44 kDa N-terminal domain for ATP-binding (NBD) [11]. Due to the highly homologous gene sequences and structural characteristics of Hsp110 and Hsp70 families, Hsp100 is also divided as Hsp70 superfamily (Hsp110/SSE) [12].

The Hsp70 genes have been identified in a number of plant species, such as rice (*Oryza sativa*) [13], *Arabidopsis thaliana* [14], spinach (*Spinacia oleracea* L.) [15], and *Gossypium raimondii* L. [16]. In *Arabidopsis thaliana*, at least 18 members of the gene family encoding Hsp70s have been identified; among these, only 14 belong to Hsp70 gene family, while the rest are HSP110/SSE family members. There are various promoter elements that respond to external stress signals, such as light response elements [17], magnetic response elements [18], and enhancer elements [19]. These promote the synthesis and maturation of the corresponding protein sequences and provide energy

by combining ATP/ADP with an ATPase domain around 44 kDa on the N-terminus, regulating the C-terminal polymorphic region of about 10 kDa and the substrate-binding region of about 18 kDa [11]. The functions of members located in the chloroplast mainly include maintaining normal operation of photosystem II, participating in the formation of thylakoid membrane, and assisting in protein translocation [20, 21]. Not all members in Hsp70s have the ability to enhance stress resistance of plants, which can be divided into constitutive and inducible types, based on their characteristics [22]. Furthermore, Hsp70s are closely related to the development of plants [23].

Tobacco (*Nicotiana tabacum*) is well known as an important economic crop and a typical model plant. High quality genome sequences of *N. tabacum* have been published and annotated thoroughly by the Solanaceae community [24]. Although the Hsp70 gene family has been studied in some plants, it has been rarely reported in tobacco. Therefore, the analysis of the distribution of Hsp70 genes in the tobacco genome, the gene structures, the promoter elements, and gene expressions would be great significance to the study of stress resistance response of tobacco during growth and development.

In this study, we performed a genome-wide survey of Sol Genomics Network databases using Hsp70 protein sequences from *Arabidopsis thaliana*. Then we used the bioinformatics methods to analyze the physicochemical properties, evolutionary relationships, conserved domains, and stress-related expression levels of the NtHsp70 gene family. This is first genome-wide analysis of NtHsp70s. Our results are relevant for the growth and development of tobacco under abiotic stress, and they provide a basis for further study of the biological functions of Hsp70 genes.

Materials and methods

Plant materials and stress treatments

The tobacco cultivar K326 (*Nicotiana tabacum* L., cv. Kentucky 326) was used for gene expression level related experiments. The tobacco seeds were sown in mixed soil (vermiculite:humus = 1:1) saturated with water in sieve-like plates. Seedlings were germinated and cultured in a greenhouse at 22 °C with a 16 h/8 h (light/dark) photoperiod for 8 weeks. Then, they were assigned to a treatment group and separately stressed by exposure to a 50 µM ABA spray, a PEG 6000 solution (−0.5 MPa), a 300 mM NaCl solution, a low temperature (4 °C), and a high temperature (42 °C), all of which were validated to cause a significant stress in pilot experiments. Untreated control plants were cultured normally. All the true leaves were sampled at 0, 1, 3, 6, 12, and 24 h after treatment. The main midribs were then

removed. After collection, all samples were quickly frozen in liquid nitrogen and stored at -80°C for RNA isolation and analysis.

Identification of *NtHsp70* genes

The Hsp70 sequences of *Oryza sativa* and *Arabidopsis thaliana* were downloaded from the Rice Genome Annotation Project (<http://rice.plantbiology.msu.edu/>) and the TAIR databases (<http://www.arabidopsis.org/>). Previously identified protein sequences of *A. thaliana* Hsp70 genes were used as query to perform BLAST (Evaluate 1e-10) searched against tobacco genome sequences to obtain the candidate dataset of tobacco Hsp70 (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). Redundant sequences were removed. Hsp70 proteins from tobacco were predicted using the hidden Markov model (HMM) profile obtained from Pfam (<http://pfam.xfam.org>, PF00012) [25] and used as queries to search the Hsp70 proteins from tobacco sequences with HMMER software (<http://hmmerr.org/>). In order to further confirm and delete uncertain Hsp70 proteins, the predicted Hsp70 domains were detected with the SMART tool (<http://smart.embl-heidelberg.de/>). Then, we obtained the final protein sequences of NtHsp70s. The biophysical properties of coding Hsp70s were calculated using the ExPASy ProtParam tool (<http://us.expasy.org/tools/protparam.html>). The subcellular localization of NtHsp70s was analyzed using ProtComp (<http://www.softberry.com/>) and WoLF PSORT II (<http://www.genscript.com/wolf-psort.html>).

Chromosome distribution and synteny analysis

The distribution information of NtHsp70 gene family in the chromosome was obtained from the Sol Genomics Network (<https://solgenomics.net/>) database. For syntenic analysis, syntenic blocks containing Hsp70 genes in the tobacco genome were identified using MCScanX program [26]. The chromosome distribution for each *NtHsp70* and synteny relationship were displayed with circos (<http://circos.ca/>) [27].

Phylogenetic analysis

The amino acid sequences of the Hsp70 genes of *A. thaliana*, *O. sativa* and *N. tabacum* were saved as FASTA format files, and multiple sequence alignments of Hsp70 genes were performed by using the ClustalX program with default parameters [28]. An unrooted phylogenetic tree was constructed using Molecular Evolutionary Genetics Analysis (MEGA) version 7.0 with the neighbor-joining method [29]. Support for the tree topology was evaluated by using a bootstrap analysis with 1000 replicates.

Gene structure and conserved motifs analysis

A diagrammatic sketch of the Hsp70 gene structure was constructed using the Gene Structure Display Server (GSDS) (<http://gsds.cbi.pku.edu.cn/>); it was based on the alignment of the cDNAs with their corresponding genomic DNA sequences. The conserved motif of the full length of Hsp70 family proteins was analyzed using the online MEME tool (Multiple Expectation Maximization for Motif Elicitation, <http://meme-suite.org/tools/meme>). The maximum motif search value was set at 10.

Promoter analysis

For cis-regulatory element analysis, 1500-bp sequences upstream from translational start sites of the putative Hsp70 genes were set as promoter sequences. The cis-regulatory elements were predicted by the PlantCARE tool (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) based on these promoter sequences.

Quantitative RT-PCR expression analysis of *NtHsp70s*

Total RNA was extracted from the sample of using a modified CTAB method. After removing genomic DNA contamination by DNase I (Fermentas, Waltham, MA, USA), 1 μg of total RNA was reverse-transcribed to cDNA using the PrimeScriptTM RT Reagent Kit (Takara, Dalian, China). The PCR amplifications were performed using LightCycler[®] 480 II (Roche Diagnostics, Indianapolis, IN, USA). Gene-specific primers were designed using Primer 6.0 (Table S1). The transcription of tobacco ribosomal protein gene L25 (GenBank accession number L18908) was used as an internal reference gene. The qRT-PCR reactions were performed on the ABI 7900 HT Real-Time PCR System (Applied Biosystems) using the following thermal cycle: 95°C for 5 min, followed by 40 cycles at 95°C for 10 s, and 60°C for 30 s. Three biological replicates were used for each gene. The relative expression level for each of *NtHsp70s* was calculated using the $2^{-\Delta\Delta\text{CT}}$ method [30].

Results

Identification of the Hsp70 genes in the tobacco genome

For the genome-wide identification of Hsp70 genes in tobacco, a local BLASTP was used to identify Hsp70 members in the tobacco genome using the Hsp70 protein sequence from *A. thaliana* as a query sequence. To verify our results, the Hsp70 domain (PF00012) was employed to

search against these candidate genes using the HMMER 3.0 program with the default E-value. By extraction, screening and identification, there were 61 protein members of the Hsp70s family in tobacco. For the nomenclature of NtHsp70, they were named as NtHsp70-1 through NtHsp70-61 (Table 1), following the previous rules for *A. thaliana*. NtHsp70 characteristics including chromosomal localization, exon number, gene length, number of amino acids, molecular mass, CDS, subcellular localization, and instability index are listed in Table 1. The results showed that the biophysical properties of different coding NtHsp70s were diverse. They were unevenly distributed on the 18 chromosomes that comprise the tobacco genome. The length of coding NtHsp70 sequences ranged from 357 to 2715 bp. More than half of the coding sequence lengths were about 2000 bp (1900 bp to 2100 bp). NtHsp70 protein sequences ranged from 118 to 904 amino acids. The molecular weights of identified NtHsp70 ranged from 12.58 to 99.88 kDa. Using the WoLF PSORT tool for subcellular localization analysis, detected NtHsp70 members were located in chloroplasts, mitochondria, cytoplasm, and endoplasmic reticulum; further, the corresponding number of members may be 4, 6, 23, and 25, respectively. However, we localized three NtHsp70 members (NtHsp70-50, NtHsp70-51, and NtHsp70-52) in unknown structures. In addition, analysis with the Expasy online tool revealed that the instability index of NtHsp70s ranged from 27.28 to 49.86. There were twelve NtHsp70 members with an instability index greater than 40, which indicated an unstable protein. The members located in the chloroplast (NtHsp70-46 to NtHsp70-49) were all stable proteins.

Chromosomal location and duplication of Hsp70 genes in tobacco

In order to clarify the distribution of Hsp70 genes on the chromosomes, the distribution map of 61 NtHsp70 genes in tobacco was analyzed based on the sequencing genome information of the Sol Genomics Network (Fig. 1). The 61 NtHsp70 genes were mapped onto 18 chromosomes and most were found distributed on both ends of the chromosome. NtHsp70 were not randomly distributed on the chromosome because of gene hot regions and gene clusters. As shown in Fig. 1, seven genes were located on chromosome 2; six on chromosome 9; five on chromosomes 15, 21 and 24; four on chromosomes 1, 4, 22 and 23; three on chromosomes 7 and 19; two on chromosomes 5, 6, 14 and 17; and one on chromosomes 8, 11 and 18. The unequal distribution of NtHsp70 genes indicated a genetic variation in the process of evolution. To research potential gene duplication within the tobacco genome, the segment duplications and tandem duplications during evolution were analyzed. Based on gene sequence homology, positional

information, and chromosomal position detection of gene duplication, the results showed that at least 15 pairs of NtHsp70 genes underwent gene duplication arising from segment duplications and tandem duplications. Among these, seven pairs of NtHsp70 genes (NtHsp70-11 and NtHsp70-60, NtHsp70-48 and NtHsp70-50, NtHsp70-18 and NtHsp70-19, NtHsp70-6 and NtHsp70-14, NtHsp70-3 and NtHsp70-47, NtHsp70-28 and NtHsp70-30, NtHsp70-2 and NtHsp70-25) belonged to tandem duplications, four (NtHsp70-1 and NtHsp70-55, NtHsp70-4 and NtHsp70-37, NtHsp70-15 and NtHsp70-20, NtHsp70-44 and NtHsp70-54) to segment duplications and four more (NtHsp70-29 and NtHsp70-40, NtHsp70-29 and NtHsp70-43, NtHsp70-43 and NtHsp70-56, NtHsp70-34 and NtHsp70-56) to gene clusters.

Phylogenetic analysis of Hsp70 genes

An unrooted phylogenetic tree was constructed for the study of the evolutionary relationship between Hsp70 members, using Molecular Evolutionary Genetics Analysis (MEGA) version 7.0 with the neighbor-joining method. Bootstrap analysis was performed with 1000 replicates for statistical reliability. There were three species in the phylogenetic tree, comprising 61 Hsp70s from tobacco, 32 Hsp70s from rice, and 18 Hsp70s from *A. thaliana*. As shown in Fig. 2, 111 Hsp70s were classified into six subfamilies (Group 1 to Group 6). Group 1 was clustered in the Hsp110/SSE subfamily. The members of Group 1 (NtHsp70-53 to NtHsp70-61) were mainly located in the endoplasmic reticulum and the cytoplasm. Group 2 to Group 6 were clustered in the DnaK subfamily. Group 2 included the largest number of NtHsp70 members (18 NtHsp70s), which were located in the most extensive subcellular regions (endoplasmic reticulum, cytoplasm, mitochondria, and unknown regions); members of Group 3 are involved in cytoplasmic metabolism (except NtHsp70-15); while members of Group 4 and Group 5 participate in metabolism occurring in the endoplasmic reticulum. Group 6 was further divided into 2 subcategories, in which Group 6-A members were located in mitochondria (except *NtHsp70-19*), while Group 6-B members belonged in chloroplasts.

Interestingly, each subfamily contained Hsp70 members from the three species: *A. thaliana*, rice, and tobacco. There were 74 Hsp70 members bearing homology relationships among 66.7% (74/111) of all members of the Hsp70s gene family. Only 2 pairs of orthologous proteins between species were identified (*NtHsp70-61* and *AtHsp70-17*; *NtHsp70-50* and *OsHsp70-3*). However, there were 35 pairs of paralogous proteins in the species.

Table 1 The information of the Hsp70 gene family in *Nicotiana tabacum*

Gene name	Gene ID	Chromosomes	Start	End	Exon number	Gene length	Number of amino acids	Molecular mass (kDa)	CDS(bp)	WoLF PSORT ^a	Instability index
NtHsp70-1	NW_015910711	Chr24	110,654,366	110,658,185	3	3819	904	99.83	2715	Endo	34.87
NtHsp70-2	NW_015899311	Chr24	110,654,473	110,658,298	3	3825	896	98.88	2691	Endo	37.17
NtHsp70-3	NW_008906668	Chr19	120,034,761	120,038,175	7	3414	668	74.48	2007	Endo	34.01
NtHsp70-4	NW_015891480	Chr21	41,190,954	41,194,731	8	3777	668	73.74	2007	Endo	29.54
NtHsp70-5	NW_015940883	Chr01	761,323	766,026	8	4703	668	73.67	2007	Endo	32.65
NtHsp70-6	NW_009543853	Chr07	14,352,043	14,355,543	7	3500	667	74.5	2004	Endo	35.56
NtHsp70-7	NW_015891199	Chr21	41,190,862	41,194,624	8	3762	667	73.98	2004	Endo	29.51
NtHsp70-8	NW_009444836	Chr21	41,190,749	41,194,753	8	4004	667	73.54	2004	Endo	27.91
NtHsp70-9	NW_015820506	Chr21	41,190,693	41,193,006	5	2313	667	73.52	2004	Endo	27.91
NtHsp70-10	NW_015903878	Chr21	41,190,758	41,194,787	8	4029	667	73.48	2004	Endo	28.32
NtHsp70-11	NW_009520113	Chr01	761,437	765,998	8	4561	666	73.44	2001	Endo	31.61
NtHsp70-12	NW_009553587	Chr15	113,938,237	113,942,792	8	4555	666	73.41	2001	Endo	29.44
NtHsp70-13	NW_015902121	Chr15	113,938,560	113,942,799	8	4239	666	73.41	2001	Endo	28.55
NtHsp70-14	NW_015855455	Chr07	14,352,732	14,356,284	7	3552	649	72.6	1950	Endo	35.81
NtHsp70-15	NW_015886411	Chr09	96,335,681	96,338,847	2	3166	648	70.88	1947	Endo	33.77
NtHsp70-16	NW_008962707	Chr17	35,644,444	35,647,715	2	3271	639	70.83	1920	Endo	34.6
NtHsp70-17	NW_015890058	Chr14	64,376,792	64,379,861	1	3069	573	62.54	1722	Endo	42.26
NtHsp70-18	NW_015808043	Chr07	63,709,136	63,712,205	1	3069	573	62.6	1722	Endo	41.18
NtHsp70-19	NW_015803591	Chr01	761,430	766,016	8	4586	290	32.03	873	Endo	33.95
NtHsp70-20	NW_015931887	Chr15	113,938,230	113,942,799	8	4569	168	18.84	507	Endo	35.9
NtHsp70-21	NW_015938689	Chr04	16,574,685	16,576,926	1	2241	659	72.41	1980	Cyto	33.35
NtHsp70-22	NW_015842167	Chr04	16,575,087	16,577,307	1	2220	656	72.06	1971	Cyto	31.72
NtHsp70-23	NW_008899791	Chr04	16,574,572	16,576,954	1	2382	654	71.88	1965	Cyto	30.46
NtHsp70-24	AY372070	Chr02	12,282,373	12,285,690	1	3317	653	71.29	1962	Cyto	35.57
NtHsp70-25	AY372069	Chr02	12,282,983	12,286,030	3	3047	653	71.01	1962	Cyto	34.28
NtHsp70-26	NW_008861535	Chr09	96,220,948	96,224,435	2	3487	651	71.11	1956	Cyto	33.28
NtHsp70-27	NW_015833318	Chr09	96,221,165	96,224,478	2	3313	651	71.11	1956	Cyto	33.81
NtHsp70-28	NW_009588212	Chr23	120,809,341	120,812,607	2	3266	650	71.28	1953	Cyto	33.92
NtHsp70-29	NW_015924923	Chr02	12,281,505	12,284,911	2	3406	649	71.1	1950	Cyto	35.03
NtHsp70-30	NW_008866702	Chr23	120,809,444	120,812,680	2	3236	648	71.09	1947	Cyto	33.12
NtHsp70-31	NW_009421259	Chr05	10,283,928	10,287,476	2	3548	648	70.97	1947	Cyto	33.56
NtHsp70-32	NW_008951052	Chr24	31,202,436	31,205,677	2	3241	648	70.97	1947	Cyto	33.7
NtHsp70-33	NW_009582398	Chr09	96,334,773	96,337,874	2	3101	648	70.97	1947	Cyto	33.84
NtHsp70-34	NW_015937266	Chr22	48,208,194	48,211,012	2	2818	585	64.35	1758	Cyto	33.99
NtHsp70-35	NW_015927047	Chr04	16,574,576	16,576,979	1	2403	573	63.01	1722	Cyto	30.88

Table 1 (continued)

Gene name	Gene ID	Chromosomes	Start	End	Exon number	Gene length	Number of amino acids	Molecular mass (kDa)	CDS(bp)	WoLF PSORT ^a	Instability index
NtHsp70-36	NW_015911947	Chr18	68,908,802	68,916,410	2	7608	187	20.97	564	Cyto	49.86
NtHsp70-37	NW_009404025	Chr09	96,223,826	96,224,278	1	452	150	16.43	453	Cyto	38.59
NtHsp70-38	NW_015907408	Chr24	32,237,282	32,237,782	2	500	145	16.62	438	Cyto	46.33
NtHsp70-39	NW_015954277	Chr24	32,237,285	32,237,715	1	430	143	15.99	432	Cyto	47.58
NtHsp70-40	NW_015901173	Chr22	19,998,284	20,002,242	6	3958	681	73.13	2046	Mito	38.72
NtHsp70-41	NW_015899916	Chr09	95,675,412	95,679,906	6	4494	681	72.99	2046	Mito	38.73
NtHsp70-42	NW_015917892	Chr02	19,998,288	20,000,811	5	2523	396	42.88	1191	Mito	41.9
NtHsp70-43	NW_015905136	Chr22	19,998,246	20,001,406	5	3160	385	41.59	1158	Mito	38.09
NtHsp70-44	NW_015882904	Chr23	126,543,515	126,545,184	5	1669	286	30.56	861	Mito	39.97
NtHsp70-45	NW_015899916	Chr14	102,300,371	102,301,059	1	688	118	12.58	357	Mito	36.71
NtHsp70-46	NW_008856380	Chr19	132,517,255	132,521,922	8	4667	707	75.31	2124	Chlo	28.8
NtHsp70-47	NW_008891824	Chr19	111,097,589	111,101,970	8	4381	707	75.17	2124	Chlo	28.92
NtHsp70-48	NW_015938024	Chr06	2,022,838	2,028,587	8	5749	692	74.45	2079	Chlo	27.28
NtHsp70-49	NW_015905563	Chr17	6,201,965	6,208,159	8	6194	689	73.93	2070	Chlo	27.78
NtHsp70-50	NW_015935569	Chr06	102,532,669	102,536,286	5	3617	582	67.02	1749	–	34.81
NtHsp70-51	NW_015807227	Chr11	38,343,217	38,345,442	4	2225	271	29.56	816	–	34.64
NtHsp70-52	NW_009452803	Chr05	31,203,464	31,207,928	4	4464	254	28.31	765	–	29.83
NtHsp70-53	NW_015802394	Chr15	11,151,589	11,157,894	10	6305	856	94.5	2571	Cyto	43.81
NtHsp70-54	NW_015910628	Chr15	11,180,281	11,186,087	9	5806	854	94.24	2565	Cyto	42.38
NtHsp70-55	NW_009518110	Chr08	1,858,770	1,864,630	9	5860	846	93.78	2541	Cyto	38.73
NtHsp70-56	NW_015794811	Chr22	20,880,101	20,885,900	9	5799	844	93.26	2535	Cyto	40.12
NtHsp70-57	NW_015903469	Chr01	103,035,522	103,050,532	9	15,010	753	84.11	2262	Cyto	44.86
NtHsp70-58	NW_015885486	Chr23	123,915,822	123,929,548	10	13,726	611	68.43	1836	Cyto	43.57
NtHsp70-59	NW_015952690	Chr02	102,965,209	102,974,730	14	9521	895	99.43	2688	Endo	39.92
NtHsp70-60	NW_015907200	Chr02	102,968,776	102,969,523	13	747	691	77.2	2076	Endo	43.57
NtHsp70-61	NW_015791011	Chr02	102,973,117	102,974,938	2	1821	330	36.33	993	Endo	33.02

Chlo chloroplast, Cyto cytosol, Endo endoplasmic reticulum, Mito mitochondria

^a Best possible subcellular localization of *Nicotiana tabacum* Hsp70 based on WoLF PSORT

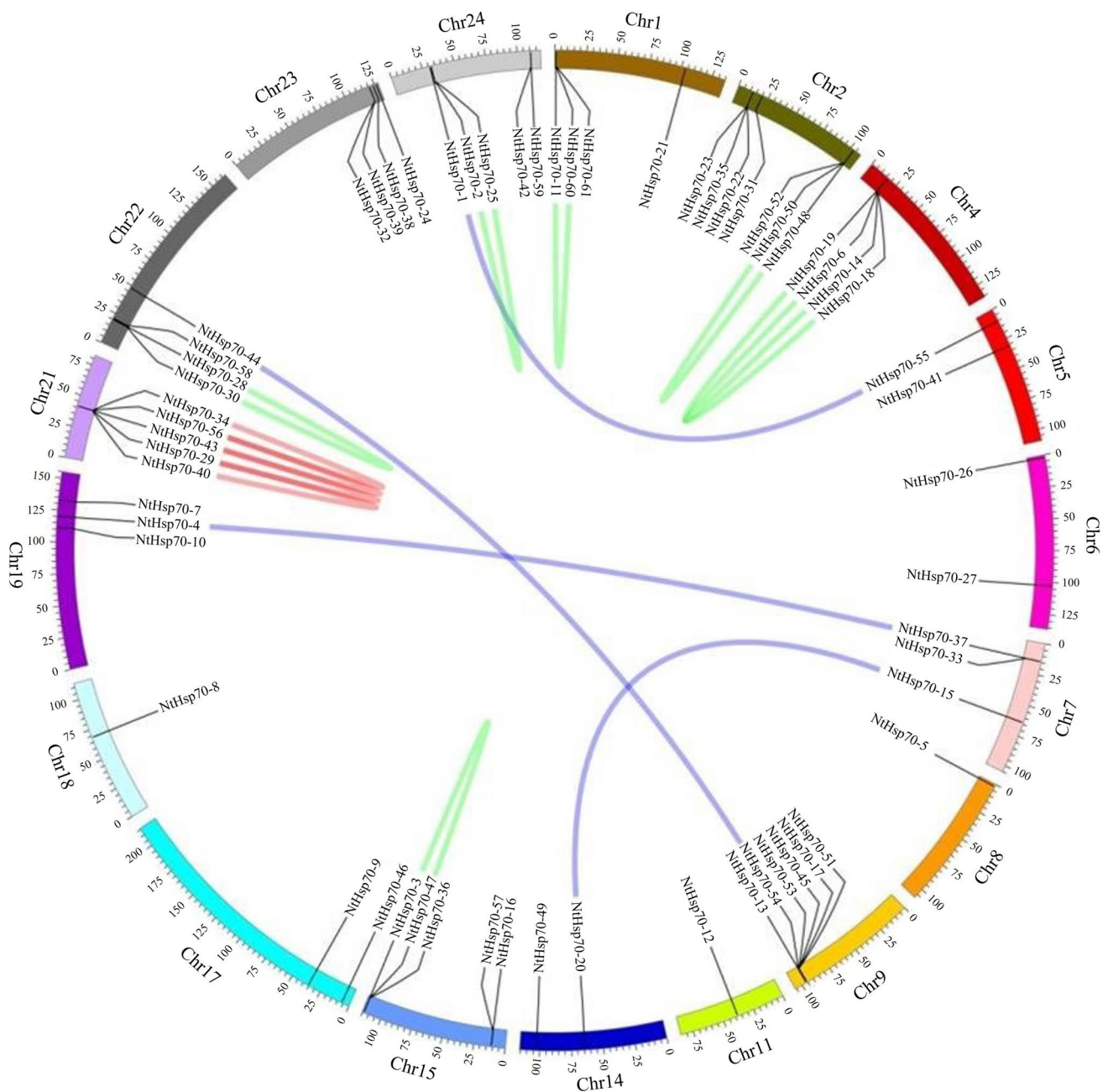


Fig. 1 Chromosome location of Hsp70 genes in *Nicotiana tabacum* genome. The chromosome distributions of NtHsp70 genes and synteny relationships were displayed using Circos software. Red lines

showed gene cluster, green lines showed tandem duplication, and blue lines showed segmental duplication. (Color figure online)

Exon–intron organization of *NtHsp70* genes

Based on information of cDNA and genomic DNA sequences, a diagrammatic sketch was constructed using the Gene Structure Display Server to analyze the structure of NtHsp70 genes. Analysis of exon–intron structure can provide important insights into the evolution of gene families [31]. A neighbor-joining phylogenetic tree was also constructed to explore whether the exon–intron distribution

patterns correlate with the phylogenetic classification. As shown in Fig. 3a, the 61 *NtHsp70* gene copies could be clearly divided into 17 categories (*NtHsp70-A* to *NtHsp70-Q*) based on the sequence divergence. Gene structure analyses indicated that the most closely related *NtHsp70* genes had a similar exon–intron distribution pattern (Fig. 3b). It is worth mentioning that, although the number of introns in different NtHsp70 family members varied widely (0 to 13 introns), the difference between members of the same subfamily was

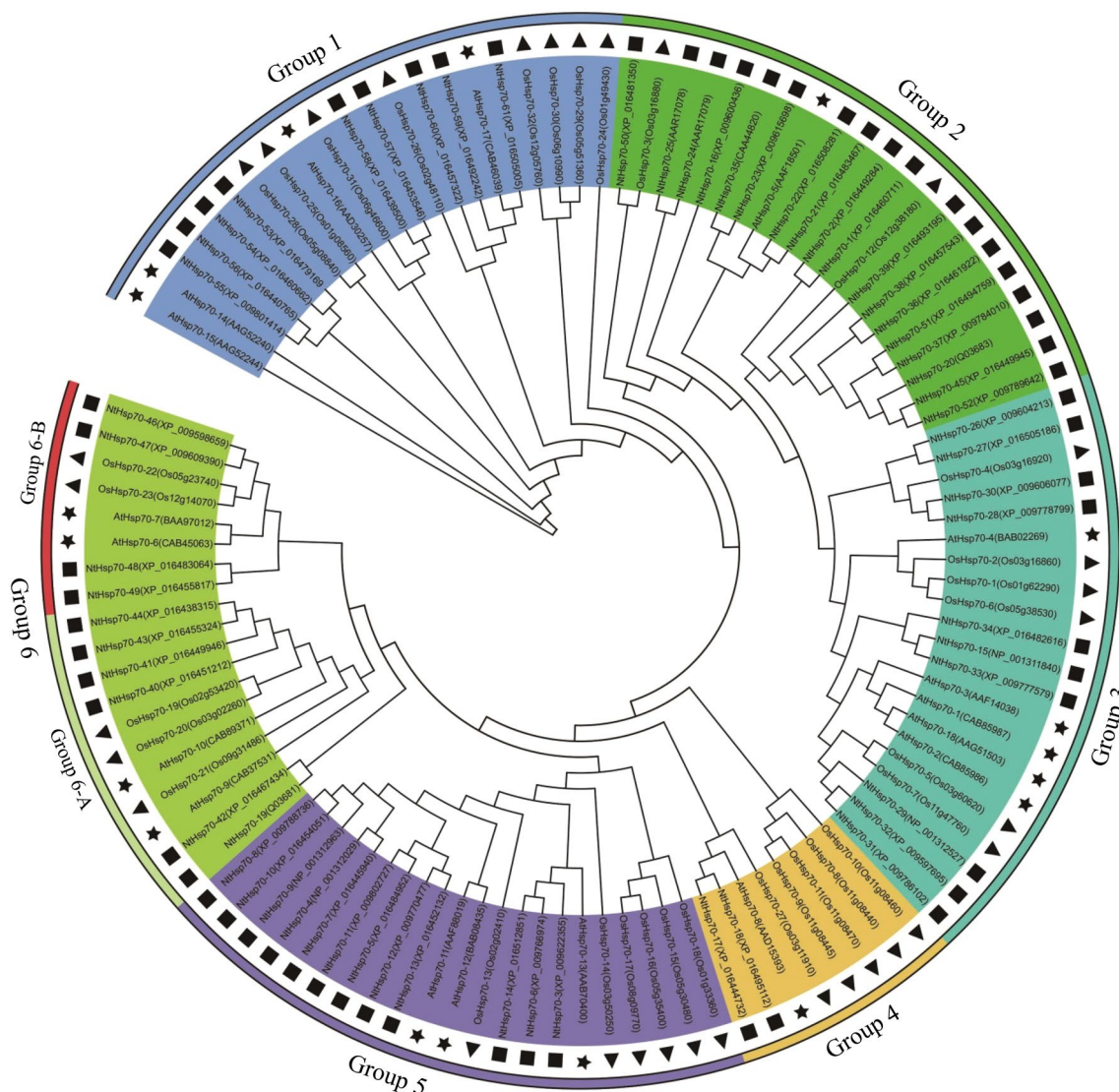


Fig. 2 Phylogenetic analysis of Hsp70 proteins from *Arabidopsis thaliana*, *Oryza sativa*, and *Nicotiana tabacum*. The unrooted phylogenetic tree was generated using MEGA7.0 program by neigh-

bor-joining method with 1000 bootstrap replicates. The rectangle represented *N. tabacum*, the triangle represented *O. sativa*, and the pentagon represented *A. thaliana*. (Color figure online)

small. These results provide significant evidence in support of the reliability of the gene structure analysis performed here. The genetic structure of members in Group 3, Group 4 and Group 6-B subfamilies was better, and the number of exons was 2, 1 and 8, respectively. The consistency in the number of introns in members of Group 1 and Group 2 was substandard, mainly due to the instability of intron numbers, which were between 1–13 and 0–7, respectively, while intron sequence-length varied greatly.

In addition, conserved motifs of NtHsp70s were analyzed using the MEME online tool. The results (Fig. 3c) showed that NtHsp70 members contained 10 conserved motifs. There were 24 NtHsp70 proteins containing all 10 conserved motifs, mainly in Group 3 and Group 5. Protein

motifs of members in Group 4 and Group 6-B were the most highly conserved. There were some differences in protein motifs of members in Group 2, due to the fluctuation of the number of motifs between 0 and 10 and the change of the position of the conserved sequence. NtHsp70-50 only contained one conserved motif, which indicated that there might be a deletion in this protein sequence. Motif 2 and motif 5 were widely distributed, their presence being confirmed in 88.5% (54/61) of NtHsp70 members. Motif-domain analysis showed that motif 2 and motif 5 together constituted the functional domain of N-terminal hexokinase. The other 8 motifs constituted the Hsp70 conservative domain. The types and distribution of conserved motifs may be related to the diversity of their gene functions.

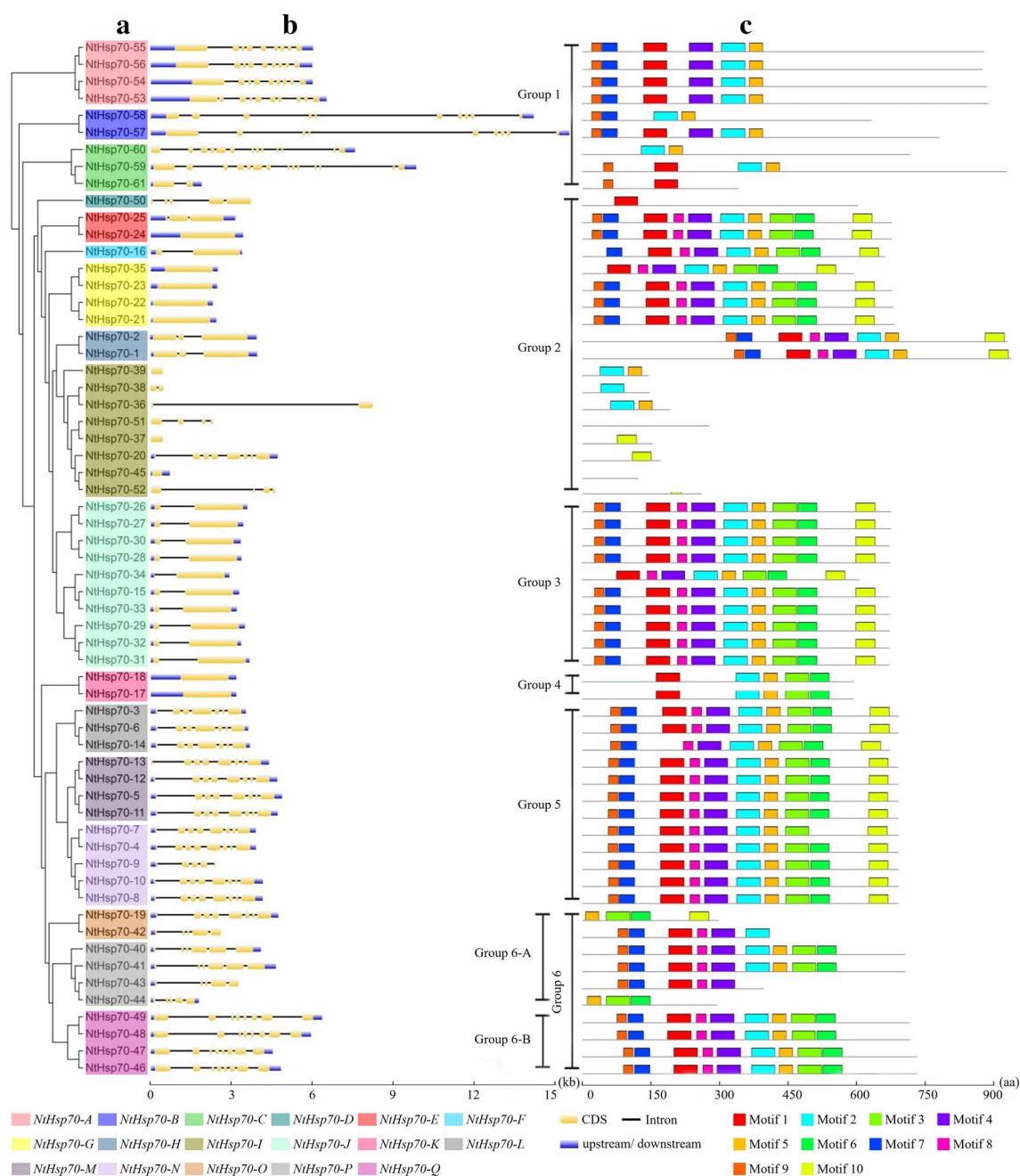


Fig. 3 Phylogenetic tree, exon–intron structure, and motif analysis of NtHsp70s. **a** Phylogenetic analysis of NtHsp70s using by neighbor-joining method. Details of clusters were shown in different color boxes. The boxes of different colors represented *NtHsp70-A* to *NtHsp70-Q* from top to bottom. **b** Gene structure analysis of NtHsp70s. Exons and introns were represented by yellow rectangles and

black lines, respectively. The untranslated regions (UTRs) were represented by blue rectangles. **c** Conserved motifs analysis of NtHsp70 proteins using MEME tools. The order of the motifs corresponded to their position in the protein sequence. Conserved motifs were showed in different colored boxes. (Color figure online)

Cis-regulatory element analysis of the NtHsp70 gene family

In order to further identify the cis-regulatory elements located upstream of NtHsp70 genes, 1500-bp sequences upstream from translational start sites of putative NtHsp70

gene families were analyzed using the PlantCARE tool. As shown in Table 2, no fewer than 16 cis-regulatory elements were identified in the promoters of the NtHsp70 genes. There were mainly three types of cis-regulatory elements, including hormone response elements, stress

Table 2 Cis-regulatory element analysis of the *NtHsp70* genes

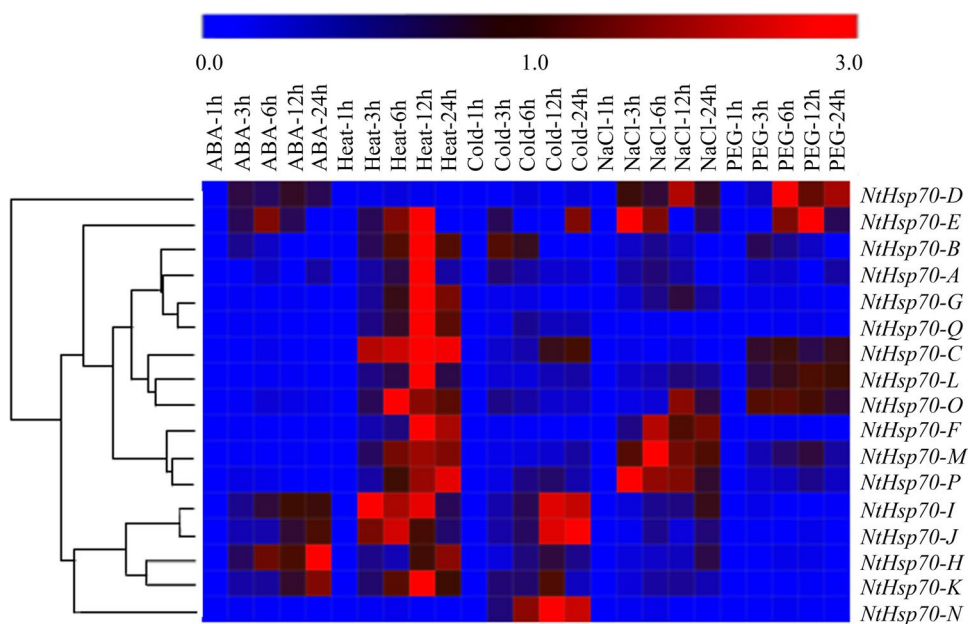
Type	Site name	Function	<i>NtHsp70</i> genes
Hormone response element	GARE-motif	Gibberellin responsive element	<i>NtHsp70-4</i> , <i>NtHsp70-5</i> , <i>NtHsp70-18</i> , <i>NtHsp70-24</i> , <i>NtHsp70-26</i> , <i>NtHsp70-27</i> , <i>NtHsp70-29</i> , <i>NtHsp70-53</i> , <i>NtHsp70-54</i> , <i>NtHsp70-55</i> , <i>NtHsp70-56</i> , <i>NtHsp70-57</i> , <i>NtHsp70-58</i>
	TGA-element	Auxin induced response element	<i>NtHsp70-3</i> , <i>NtHsp70-4</i> , <i>NtHsp70-7</i> , <i>NtHsp70-41</i> , <i>NtHsp70-43</i>
	ERE	Ethylene induced response element	<i>NtHsp70-26</i> , <i>NtHsp70-56</i> , <i>NtHsp70-57</i>
	ABRE	ABA-responsive element	<i>NtHsp70-6</i> , <i>NtHsp70-11</i> , <i>NtHsp70-14</i> , <i>NtHsp70-18</i> , <i>NtHsp70-19</i> , <i>NtHsp70-35</i> , <i>NtHsp70-54</i> , <i>NtHsp70-55</i> , <i>NtHsp70-56</i> , <i>NtHsp70-58</i>
	TCA-element	Salicylic acid responsive element	<i>NtHsp70-17</i> , <i>NtHsp70-18</i> , <i>NtHsp70-23</i> , <i>NtHsp70-35</i> , <i>NtHsp70-45</i>
Stress-induced component	TC-rich repeats	Defense and stress response element	<i>NtHsp70-3</i> , <i>NtHsp70-17</i> , <i>NtHsp70-18</i> , <i>NtHsp70-27</i> , <i>NtHsp70-30</i> , <i>NtHsp70-31</i> , <i>NtHsp70-48</i> , <i>NtHsp70-49</i> , <i>NtHsp70-55</i> , <i>NtHsp70-56</i>
	HSE	Heat shock element	<i>NtHsp70-30</i> , <i>NtHsp70-35</i> , <i>NtHsp70-54</i> , <i>NtHsp70-55</i> , <i>NtHsp70-56</i> , <i>NtHsp70-57</i> , <i>NtHsp70-58</i>
	LTR	Low-temperature responsive element	<i>NtHsp70-57</i> , <i>NtHsp70-58</i>
	ARE	Hypoxia inducible response element	<i>NtHsp70-3</i> , <i>NtHsp70-6</i> , <i>NtHsp70-14</i> , <i>NtHsp70-16</i> , <i>NtHsp70-18</i> , <i>NtHsp70-23</i> , <i>NtHsp70-35</i> , <i>NtHsp70-54</i> , <i>NtHsp70-55</i> , <i>NtHsp70-56</i>
	WUN-motif	Wound induction response element	<i>NtHsp70-18</i> , <i>NtHsp70-56</i> , <i>NtHsp70-59</i>
	Box-W1	Fungal pathogen response element	<i>NtHsp70-54</i> , <i>NtHsp70-56</i>
	MBS	Drought stress response element	<i>NtHsp70-5</i> , <i>NtHsp70-11</i> , <i>NtHsp70-12</i> , <i>NtHsp70-17</i> , <i>NtHsp70-19</i> , <i>NtHsp70-20</i> , <i>NtHsp70-24</i> , <i>NtHsp70-25</i> , <i>NtHsp70-35</i> , <i>NtHsp70-53</i> , <i>NtHsp70-56</i>
Light control element	Circadian	Day-and-night induction response element	<i>NtHsp70-16</i> , <i>NtHsp70-17</i> , <i>NtHsp70-18</i> , <i>NtHsp70-22</i> , <i>NtHsp70-24</i> , <i>NtHsp70-25</i> , <i>NtHsp70-57</i>
	G-box	Light-induced response element	<i>NtHsp70-5</i> , <i>NtHsp70-8</i> , <i>NtHsp70-9</i> , <i>NtHsp70-10</i> , <i>NtHsp70-11</i> , <i>NtHsp70-16</i> , <i>NtHsp70-17</i> , <i>NtHsp70-18</i> , <i>NtHsp70-19</i> , <i>NtHsp70-23</i> , <i>NtHsp70-24</i> , <i>NtHsp70-35</i> , <i>NtHsp70-53</i> , <i>NtHsp70-54</i> , <i>NtHsp70-55</i> , <i>NtHsp70-56</i> , <i>NtHsp70-57</i> , <i>NtHsp70-58</i>
	Box I	Light-induced response element	<i>NtHsp70-17</i> , <i>NtHsp70-26</i> , <i>NtHsp70-56</i> , <i>NtHsp70-57</i> , <i>NtHsp70-58</i>
	3-AF1 binding site	Light-induced response element	<i>NtHsp70-32</i> , <i>NtHsp70-47</i>

response-related reactive elements and light-induced reactive-related elements. The main components associated with abiotic stress responses were a defense and stress response element (TC-rich repeats), the heat shock element (HSE), the low-temperature responsive element (LTR), the hypoxia inducible response element (ARE), the wound-induction response element (WUN-motif), and the drought-stress response element (MBS). These abiotic response elements indicate that *NtHsp70* genes have a wide participation in plant stress responses, thereby improving the chances for organisms to escape or better cope with the damaging effects of adverse environmental conditions.

Expression profiles of *NtHsp70* genes in response to abiotic stress

Over the long-term evolutionary history, plants have developed a series of mechanisms to protect themselves from stress. In order to further clarify the potential functions of *NtHsp70* genes in abiotic stress responses, the expression of 17 *NtHsp70s* subclasses under ABA, drought, salinity, cold, and heat conditions was analyzed by qRT-PCR. Expression analysis showed that almost all *NtHsp70* genes could be induced by various abiotic stresses, though had significant spatial differences (Fig. 4). *NtHsp70-A*, *NtHsp70-B*, *NtHsp70-D*, *NtHsp70-E*, *NtHsp70-H*, *NtHsp70-I*, *NtHsp70-J*, and *NtHsp70-K* were induced by

Fig. 4 Expression profiles of *NtHsp70s* responding to ABA, PEG, NaCl, low and high temperature treatments. The data from quantitative RT-PCR was used for hierarchical cluster analysis with MeV 4.9.0. The expression values of each *NtHsp70* were log₂ transformed. Blue color showed low level expression and red represented high level expression. (Color figure online)



ABA treatment. Except for *NtHsp70-D* and *NtHsp70-N*, all *NtHsp70* genes were induced by heat treatment. *NtHsp70-A*, *NtHsp70-B*, *NtHsp70-E*, *NtHsp70-H*, *NtHsp70-I*, *NtHsp70-J*, *NtHsp70-K*, *NtHsp70-M*, *NtHsp70-O*, and *NtHsp70-P* were all induced by cold and NaCl treatments. The transcripts of *NtHsp70-A*, *NtHsp70-B*, *NtHsp70-C*, *NtHsp70-D*, *NtHsp70-E*, *NtHsp70-L*, *NtHsp70-M*, *NtHsp70-O*, and *NtHsp70-P* were induced by PEG treatment. Interestingly, among the five abiotic stress treatments tested, heat stress induced changes in transcript abundance of most *NtHsp70* genes, but particularly in abundance of *NtHsp70-D* and *NtHsp70-N* transcripts, which were induced to a significantly greater extent than they were by ABA, drought, salinity, or cold shock. The expression of these *NtHsp70* genes peaked at 12 h in response to heat treatment. The expression profile indicated that *NtHsp70* genes might be involved in tobacco responses to abiotic stress, especially to high temperature stress.

Discussion

Hsp70 is a widespread family of molecular chaperones distributed in all eukaryotic cells [8]. Identification and analysis of Hsp70 genes in tobacco will provide value information for researches interested in stress resistance in other plant species. In this study, 61 Hsp70 genes were detected in tobacco. Among them, nine belonged to the Hsp70 superfamily (Hsp110/SSE). Compared to other analyzed plant Hsp70 gene families (32 Hsp70 genes have been identified in *O. sativa*, 30 in *G. raimondii*, and 18 in *A. thaliana*), the *NtHsp70* family is the largest, with 61 members. The

reason for this may be that tobacco is an allotetraploid species that has undergone extensive genetic replication; thus, the physicochemical properties of different *NtHsp70s* are different. These results may be related to the diversity of cell functions. The instability index reflects the stability of proteins [32]. *NtHsp70s* located in chloroplasts were all stable proteins, which may be related to their participation in maintaining normal operation of photosystem II [20]. *NtHsp70* genes were unevenly distributed on chromosomes, mainly on both ends of the chromosome (Fig. 1), much as the Hsp70 genes of *A. thaliana*. Gene duplication is an important mechanism of evolution of gene families [33]. We identified at least 15 pairs of *NtHsp70* genes that have undergone gene duplication, through segment and tandem duplications. Not only does duplication increase the number of family members; in addition, tandem and segment duplications may, to a certain extent, alleviate function loss due to genetic or chromosomal mutation. Further, they are believed to play a role in the specificity and diversity of gene function.

Phylogenetic trees help in analyzing the evolutionary relationships between genes. We constructed a phylogenetic tree with 111 Hsp70s from rice, *A. thaliana*, and tobacco (Fig. 2). *NtHsp70s* proteins were divided into 6 groups, which were similar to those reported by a previous study of Hsp70s in *A. thaliana*. Among them, each group contains Hsp70s from *A. thaliana*, *O. sativa*, and *N. tabacum*, indicating that the *NtHsp70* family genes were earlier product of plant evolution than the divergence of dicots and monocots. In addition, the phylogenetic tree showed that there were 2 pairs of orthologous genes and 35 pairs of paralogous genes, implying that most species expanded according to their own species-specific approach during the evolution of Hsp70s. This

phenomenon has been reported in other studies of plant gene families [34]. Gene structure and conserved domain analysis indicated that *NtHsp70* genes in the same group had similar conserved motifs. These conserved motifs played an important role in the function of specific groups. As is well known, protein structure determines function [35]. From the point of view of protein primary structure, amino acid sequences can provide important information on evolutionary relationships. We found that the members of Group 2 were more complex, the genetic structures were less consistent, and the protein sequences were the least conservative. This may be due to the rapid differentiation of the gene structure and functions within the *NtHsp70* gene family in the period when Group 2 appeared. In addition to the common conserved motifs, most *NtHsp70* members contained N-terminal hexokinase conserved motifs, which helped Hsp70 interact with ATP/ADP and adjusted the hinge region [36].

Like the conserved motifs of proteins, promoter sequences also play a key role in the divergence of Hsp70 gene functions. Cis-regulatory element analysis showed that most tobacco Hsp70 gene promoter sequences (40/61 *NtHsp70* genes) contained cis-regulatory elements associated with abiotic stress (Table 2). An HSE element has been proved a contributing factor of plant response to heat stress [37]. However, there are many cis-acting elements that have not been identified. There were mainly three types of cis-regulatory elements in tobacco, including hormone response elements, stress response-related reactive elements and light-induced reactive-related elements. For example, ETH mediates transmission of wound signals to prevent tissue necrosis [38]; ABA participates in metabolic regulation of drought stress [39]; SA plays an important role in invasion of pathogenic microorganisms and enhances tobacco resistance to mosaic disease [40]. These results indicate that tobacco Hsp70s are extensively involved in response to biotic and abiotic stress and provide a theoretical basis for further analysis of the mechanism of action of tobacco Hsp70 genes. However, in this study, there was no clear correlation between cis-regulatory elements of the promoter region of *NtHsp70* genes and their number or gene expression. There may be three reasons for this phenomenon. First, it may be related to the spatiotemporal specificity of gene expression. Secondly, many unidentified cis-regulatory elements were detected, which might be ignored. Finally, some sequence diversity existed in these cis-regulatory elements, and the genes they regulate produce different functions, which usually causes different plant phenotypes.

Numerous studies have shown that Hsp70s participate in a variety of abiotic stress responses [41]. Eighteen Hsp70 gene members were identified in *A. thaliana*. These genes exhibit distinct expression profiles under heat stress [14]. Overexpression of *AtHsp70-1* could improve the heat and salt tolerance of *A. thaliana* [42]. Hsp70 (BiP) gene of the

endoplasmic reticulum in spinach was highly expressed under cold stress [43]. Similarly, many Hsp70 genes were identified in *S. tuberosum* (20) and *C. annuum* (21). Transcript expression analysis indicated that these Hsp70 genes could be remarkably induced by abiotic and hormone stresses [9, 44]. As a natural allopolyploid, there were 61 gene copies of the Hsp70 gene family existed in the genome of *N. tabacum*. These *NtHsp70* copies were divided into 17 categories (gene members) base on the sequence divergence of the predicted coding region. Our results indicated that 17 *NtHsp70* gene members were involved in at least one or more abiotic stress responses, highlighting the wide participation of *NtHsp70* genes in environmental adaptation. The expression patterns of *NtHsp70* genes were different under different abiotic stress conditions. Expression of almost all *NtHsp70* genes increased significantly under heat stress treatment and peaked at 12 h, including 88.2% (15/17). This may suggest that *NtHsp70* genes are more sensitive to heat stress. Although not all *NtHsp70s* contained HSE cis-acting elements, the lack of a specific element does not necessarily imply that the genes were “uninduced”.

Absciscic acid (ABA) is an essential plant hormone that is involved in the abiotic stress response. Numerous evidence shows that ABA acts through regulatory pathways that control gene expression and stomatal closure [45, 46]. *AREB1* is leucine zipper transcription factor that binds to the absciscic acid (ABA) responsive element (ABRE) motif in the promoter region of ABA-inducible genes. *AREB1* overexpressing plants were hypersensitive to ABA, and thereby showed promoted stomatal closure under drought stress [47]. In rice, identification of gene modules associated with drought response indicated that the ABRE motif was conserved in the drought response genes. These drought response genes may be involved in the ABA-dependent drought response pathway [4]. Here, ten *NtHsp70s* contained ABRE motifs in the promoter regions, and most of these genes were activated by ABA and drought stress. In addition, some *NtHsp70s* involved in drought response were not induced by ABA treatment. Further studies are required on the relationship between *NtHsp70s* and the ABA independent signal transduction pathway.

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Authors' contributions ZPS and FLP conceived and designed the experiments. XPL and DBW performed the experiments and participated to the data analysis. CY and BQZ performed the qRT-PCR experiments. HYZ revised the manuscript.

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Data availability The data sets supporting the results of this article are included within the article and its additional file.

Compliance with ethical standards

Conflicts of interest The authors declare that they have no conflicts of interest.

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